TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Measurement of High Temperature by Thermocouple and Potentiometer

A junction of two metals in a furnace generates a few millivolts; a potentiometer measures that emf without drawing any current from it, and the emf reads off the temperature.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 16 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to measure the temperature of an electric furnace (several hundred °C, beyond any liquid-in-glass thermometer) with a copper–constantan thermocouple whose emf is measured by a potentiometer. When two junctions of dissimilar metals stand at different temperatures, a thermo-emf appears (Seebeck effect), depending only on the two temperatures and the pair of metals:

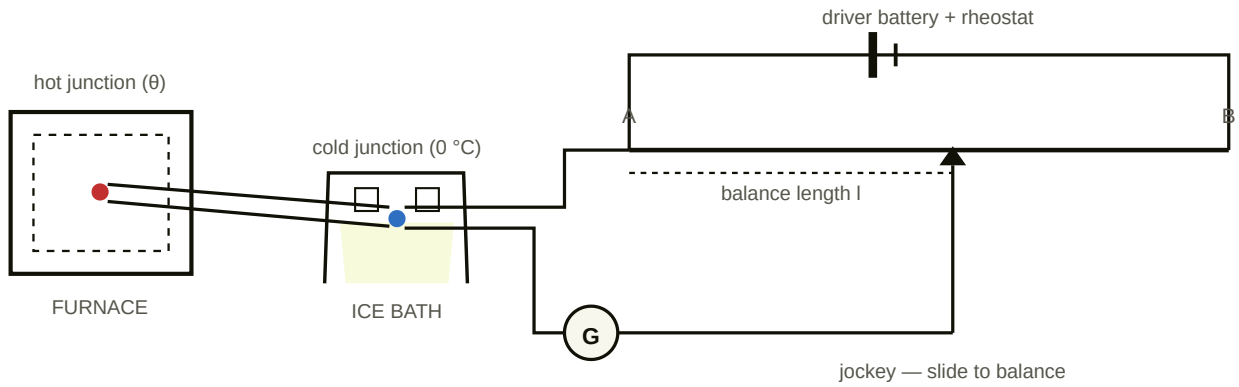
$$E \approx a\,(\theta - \theta_0) \quad \text{over a moderate range}$$

where θ is the hot-junction and θ_0 the cold-junction temperature. For copper–constantan the sensitivity a is about 41 μV °C⁻¹. The cold junction is held in melting ice (θ_0 = 0 °C), so E reads θ almost directly once the pair has been calibrated at known fixed points.

The potentiometer measures E by **null balance**: a steady potential gradient is maintained along a uniform wire, and the thermocouple is opposed against the fall of potential along a length l of that wire. At balance the galvanometer reads zero, no current leaves the thermocouple, and

$$E = (\text{potential gradient}) \times l\,, \quad \text{gradient} = E_s / l_s$$

where E_s is the emf of a standard cell balanced at length l_s. Because no current is drawn at balance, the potentiometer reads the true emf — a voltmeter, which must draw current, could not.



NULL METHOD: AT BALANCE THE GALVANOMETER READS ZERO — NO CURRENT LEAVES THE THERMOCOUPLE

Figure 1 — Thermocouple and potentiometer. The hot junction sits in the furnace, the cold junction in ice; the thermo-emf is opposed against the potential fall along wire AB, whose gradient is fixed by balancing a standard cell on the same wire.

2. Apparatus and procedure

- **Apparatus:** copper–constantan thermocouple with the hot junction in a protective sheath, ice bath, electric furnace with a rough thermometer or controller, multi-wire potentiometer, standard cell (E_s = 1.0186 V), sensitive galvanometer with lamp-and-scale or centre-zero meter, driver battery, rheostat, keys and leads.
- Pack the cold junction in melting ice in a thermos or beaker; the hot junction goes into the furnace, which is switched on and allowed to settle at a steady temperature.
- Standardise the potentiometer: connect the standard cell in place of the thermocouple, set the rheostat, and find its balance length l_s. The gradient is then E_s/l_s. (The driver current must not be touched again after this.)
- Reconnect the thermocouple — positive lead to A — and slide the jockey until the galvanometer reads exactly zero. Read the balance length l. Tap the jockey only momentarily while hunting, and never drag it along the wire under pressure.
- Compute $E = (E_s/l_s) \times l$, and the temperature from the calibration: $\theta = E/a$ with $a \approx 41$ μV °C^{-1} for copper–constantan (or read θ from the calibration curve plotted at fixed points — ice, steam, and the furnace checked against a known thermometer at lower settings).
- Repeat at a second furnace setting to check consistency.

3. Worked calculation

Standard cell E_s = 1.0186 V balances at l_s = 101.86 cm:

$$\text{gradient} = 1.0186 / 101.86 = 0.01000 \text{ V cm}^{-1}$$

The thermocouple (cold junction in ice) balances at l = 2.05 cm:

$$E = 0.01000 \times 2.05 = 0.0205 \text{ V} = 20.5 \text{ mV}$$

$$\theta = 20.5 \times 10^{-3} / (41 \times 10^{-6}) = 500 \text{ °C}$$

RESULT

Furnace temperature ≈ 500 °C. Over a wide range E is not exactly proportional to θ (the calibration curve bends slightly), so precision work reads θ from the plotted curve rather than from a single sensitivity value; the linear estimate is good to a few degrees here.

4. Precautions

- The cold junction must stay at 0 °C in melting ice throughout — the thermocouple measures a *difference* of temperature, and a warming cold junction silently subtracts from the answer.
- Never alter the driver-battery rheostat between standardising and measuring; the whole measurement rests on the gradient staying exactly as the standard cell set it.
- Protect the standard cell: it must never supply current. Keep its key closed only for the instant of balancing, and never short it.
- Mind the polarity — a reversed thermocouple gives no balance anywhere on the wire. Check that the balance point lies comfortably on the wire, not jammed at an end.
- The furnace must be steady while balancing; a drifting hot junction gives a wandering null point.

Practice (viva preparation)

1. Why is the emf measured with a potentiometer rather than a sensitive voltmeter?
2. At another furnace setting the balance length is 1.64 cm with the same gradient as the worked example. Find the temperature.
3. The cold junction is left in water at 25 °C instead of ice, and the emf corresponds to 500 °C on the ice-based calibration. What is the true hot-junction temperature?
4. Why is copper–constantan unsuitable at 900 °C, and which common pair is used there instead?
5. The galvanometer cannot be brought to zero anywhere on the wire. Give two distinct possible causes.

Answer key

1. At balance the potentiometer draws no current from the thermocouple, so it measures the true emf. A voltmeter draws current, and the voltage drop inside the thermocouple's own resistance makes its terminal voltage smaller than the emf — a built-in underestimate.
2. $E = 0.01000 \times 1.64 = 16.4$ mV; $\theta = 16.4 \times 10^{-3}/(41 \times 10^{-6}) = 400$ °C.
3. The emf measures $\theta - \theta_0 = 500$ °C with $\theta_0 = 25$ °C, so $\theta = 500 + 25 = 525$ °C. Ignoring the cold-junction temperature under-reads by exactly that amount.
4. Copper oxidises rapidly and its thermoelectric behaviour degrades well below 900 °C (the pair is normally limited to about 400–500 °C); at furnace temperatures chromel–alumel (to ~1200 °C) or platinum/platinum–rhodium (to ~1500 °C) is used.
5. (i) The thermocouple is connected with reversed polarity, so its emf aids rather than opposes the wire's fall of potential; (ii) the potential gradient is too small (or the emf too large) so the balance point would lie beyond the end of the wire — or the driver battery is exhausted/misconnected so no gradient exists at all.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib